

Production Order List API — API Specification Design (API仕様設計)

002_DD-API — supports 002_DD; handlers designed in 002_DD-FN; requirements REQ-020–REQ-027.

Document control (改版履歴)

Field	Value
Document ID	002_DD-API
System name	ProductionManagementAI
Subsystem name	Production orders
Work item	WI-003
Created by	Claude (for ThanhTN)
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Version	Date	Author	Revision content
1	2026-09-22	Claude (for ThanhTN)	Initial creation
2	2026-09-23	Claude (for ThanhTN)	WI-005: the UI is Japanese. The example product name is a Japanese automobile part (WI-005 DEC-002); Problem Details <code>title</code> texts stay English (API text, never shown in the UI). No field, rule, API or processing changes

Overview and operation catalog

Field	Value
API / module name	Production order list API
Overview	One read-only JSON endpoint behind SCR-002, plus the existing product endpoint reused for the product filter. Adds no write path (DEC-003)

No	Endpoint / method	Handler	Purpose	Notes
1	GET /api/production-orders	ProductionOrdersController.List → ProductionOrderService.ListAsync	One page of orders matching the filters, with the total	New in WI-003
2	GET /api/products	ProductsController.List → ProductionOrderService.ListProductsAsync	Product filter options	Existing (001_DD-API §1), reused unchanged — 001_DD-API anticipated this

Common to both endpoints, unchanged from 001_DD-API:

- **Auth:** cookie session (0002_ADR) plus policy `ProductionOrderEditor` = role `Admin` or `Operator`. No session → 401 with an empty body; session without the role → 403 with an empty body. The policy name is kept although this endpoint only reads: introducing a second, read-only policy would imply a permission split that WI-001 DEC-015 has not decided. Recorded as a limitation in 002_DD, not invented here.
- **Content type:** responses are `application/json`; errors are `application/problem+json` (RFC 9457). The list endpoint has no request body, so the 415 rule in 001_DD-API does not apply to it.
- **JSON:** camelCase; dates `YYYY-MM-DD`; timestamps ISO 8601 UTC with `Z`; enums as strings.
- **CSRF:** not applicable — `GET`, no state change, no request body (WI-002 DEC-020 posture unchanged).
- **Error body:** the RFC 9457 shape defined in 001_DD-API, with `code` and `errors` carrying message IDs from the single catalog. Screen B's new IDs are MSG-E015–MSG-E020 and MSG-I003–MSG-I004; MSG-E002 and MSG-E013 are reused. Stack traces and exception text are never returned.

```
{
  "type": "urn:pmi:problem:validation",
  "title": "One or more query parameters are invalid.",
  "status": 400,
  "code": "VALIDATION",
  "errors": { "dueFrom": ["MSG-E017"], "pageSize": ["MSG-E019"] },
  "traceId": "00-4bf92f3577b34da6a3ce929d0e0e4736-00f067aa0ba902b7-01"
}
```

System-wide API registry entries

ID	Endpoint	Overview	Notes
API-PO-04	GET /api/production-orders	List production orders (filter, sort, page)	New; continues 001_DD-API's registry

API-PRD-01 (GET /api/products) and API-PO-01 – API-PO-03 stay as registered in 001_DD-API.

Shared response object: ProductionOrderListItem

One element of the `items` array. It is **not** `ProductionOrderResponse` (001_DD-API): the list deliberately returns a narrower row — no `notes`, no `version`, no `allowedNextStatuses`, no `isProductQuantityEditable` — because the list neither shows nor writes those, and 002_DB's projection does not read them. The two shapes are intentionally separate so a later change to the edit contract cannot silently widen the list.

No	Name	Variable name	Type	Length	Required	Repeats (array)	Value mapping	Example	Description	Notes
1	ID	<code>id</code>	string (uuid)	36	yes	no	—	3f2c...	<code>production_orders.id</code>	Row link target (REQ-025)
2	Order number	<code>orderNumber</code>	string	13	yes	no	—	P0-2026-00042	Generated column	
3	Product	<code>product</code>	object	—	yes	no	Join to <code>products</code>	—	<code>{ "id", "sku", "name" }</code>	Rendered as "{sku} — {name}" (002_BD M-06)
3.1	Product ID	<code>product.id</code>	string (uuid)	36	yes	no	—			
3.2	Product SKU	<code>product.sku</code>	string	50	yes	no	—	P-1004		
3.3	Product name	<code>product.name</code>	string	200	yes	no	—	ドライブシャフト		
4	Quantity	<code>quantity</code>	integer	—	yes	no	—	250		
5	Due date	<code>dueDate</code>	string (date)	10	yes	no	—	2026-10-01	Plant-local date, no timezone conversion	
6	Status	<code>status</code>	string	—	yes	no	Enum name	InProgress		Label mapped client-side (002_BD M-05)
7	Overdue	<code>isOverdue</code>	boolean	—	yes	no	<code>dueDate < plantToday && status ∈ {Draft, InProgress}</code>	true	Computed server-side, never stored (002_DB)	Drives M-08; the server owns "today" (WI-002 DEC-011)
8	Updated at	<code>updatedAt</code>	string (date-time)	—	yes	no	UTC	2026-09-20T01:12:03Z		

Shared response object: `PagedResult<T>`

No	Name	Variable name	Type	Required	Repeats (array)	Value mapping	Example	Description
1	Items	<code>items</code>	<code>ProductionOrderListItem[]</code>	yes	yes	—	—	The requested page; empty when the page is past the last one
2	Total	<code>total</code>	integer	yes	no	Exact <code>count(*)</code> over the same filters	<code>87</code>	Drives the summary and page count (002_BD M-09)
3	Page	<code>page</code>	integer	yes	no	Echo of the effective value	<code>1</code>	
4	Page size	<code>pageSize</code>	integer	yes	no	Echo of the effective value	<code>20</code>	
5	Sort	<code>sort</code>	string	yes	no	Echo of the effective value	<code>dueDate</code>	
6	Direction	<code>dir</code>	string	yes	no	Echo of the effective value	<code>asc</code>	

The four echoed controls are returned so the client renders from the response it actually got, not from what it believes it asked for — which keeps the URL, the table and the summary consistent even if a request is retried or arrives out of order.

Endpoint / method design

1. GET `/api/production-orders`

Field	Value
Description	Return one page of production orders matching the filters, in the requested order, with the total number of matches
Return type	<code>PagedResult<ProductionOrderListItem></code>
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Arguments: query-string parameters only — see Request fields. No route or body parameters.

Processing overview: bind the query string **as strings only** — every value is parsed by the validator, so each failure carries its designed message ID instead of a generic model-binding error — then validate it (allow-lists for `status`, `sort`, `dir`, `pageSize`; parse dates; bound `page`; trim, bound and escape the order-number

fragment) and hand a validated query object to `ProductionOrderService.ListAsync` (002_DD-FN §1), which runs the count and the page query described in 002_DB and marks each row's `isOverdue` against the plant clock. Absent parameters take their defaults; parameters present but invalid are rejected, never defaulted (002_BD V-13), so a crafted request cannot widen the query.

Processing flow

Step	Description	Calls
1	Authorize	policy <code>ProductionOrderEditor</code>
2	Bind and validate the query parameters; on failure return 400 with per-parameter message IDs	<code>ProductionOrderListQuery.TryCreate</code> (002_DD module 5)
3	Verify the product filter exists, when one was sent; on failure 400 <code>productId: MSG-E002</code>	<code>IProductionOrderRepository.ProductExistsAsync</code> (existing)
4	Count matches, then fetch the page	<code>ProductionOrderService.ListAsync</code> → 002_DD-FN §1
5	Map rows to <code>ProductionOrderListItem</code> , computing <code>isOverdue</code> from <code>IPlantClock.Today</code>	002_DD-FN §1
6	200 with <code>PagedResult</code>	—

Return value

Type	Name	Description
<code>PagedResult<ProductionOrderListItem></code>	body	See the shared objects above

Request fields

Every parameter is optional. `status` is the only repeatable one (`?status=Draft&status=InProgress`) — the multi-select filter, DEC-005.

No	Name	Variable name	Type	Length	Required	Source	Example	Description	Notes
1	Status	<code>status</code>	string (repeatable)	—	no	query	<code>Draft</code>	Restrict to the given statuses; absent = no restriction	Allow-list <code>Draft</code> <code>InProgress</code> <code>Completed</code> <code>Cancelled</code> , matched by name only — a numeric enum value such as <code>3</code> is rejected; duplicates collapse; unknown value → 400 <code>MSG-E018</code> (002_BD V-11)
2	Product	<code>productId</code>	string (uuid)	36	no	query	<code>0197e4a0-...</code>	Restrict to one product	Unparseable → 400 <code>MSG-E002</code> ; parseable but unknown → 400 <code>MSG-E002</code> (002_BD V-12)
3	Due from	<code>dueFrom</code>	string (date)	10	no	query	<code>2026-09-01</code>	Inclusive lower bound on <code>dueDate</code>	Malformed → 400 <code>MSG-E016</code>

No	Name	Variable name	Type	Length	Required	Source	Example	Description	Notes
4	Due to	dueTo	string (date)	10	no	query	2026-10-31	Inclusive upper bound on dueDate	Malformed → 400 MSG-E016 ; dueFrom > dueTo → 400 MSG-E017 on dueFrom (002_BD V-10)
5	Order number	orderNumber	string	≤ 20	no	query	00042	Case-insensitive fragment of the order number	Trimmed first; longer than 20 after trimming → 400 MSG-E015 ; empty after trimming = absent (002_BD V-09)
6	Sort	sort	string	—	no	query	dueDate	Sort key	Allow-list orderNumber product quantity dueDate status updatedAt ; default dueDate ; unknown → 400 MSG-E019
7	Direction	dir	string	—	no	query	asc	Sort direction	Allow-list asc desc ; default asc ; unknown → 400 MSG-E019
8	Page	page	integer	—	no	query	2	1-based page number	Default 1; < 1 , non-integer or > 100000 → 400 MSG-E019 . A page past the last one is valid and returns an empty items (REQ-024)
9	Page size	pageSize	integer	—	no	query	20	Rows per page	Allow-list 10 20 50 100; default 20; anything else → 400 MSG-E019 (DEC-002)

Normalization of the order-number fragment (DEC-010): trim → reject if longer than 20 → upper-case → escape \, % and _ with a \ prefix → wrap as %fragment% → match with LIKE ... ESCAPE '\ ' against order_number . Upper-casing is enough for case-insensitivity because order_number is generated and always upper-case, and it keeps the trigram index usable. The pattern is a bound parameter; nothing is concatenated into SQL (ai/rules/database.md).

Response fields: PagedResult<ProductionOrderListItem> — see the shared objects above.

Example response:

```
{
  "items": [
    {
      "id": "0197e4a0-0000-7000-8000-0000000002a1",
      "orderNumber": "PO-2026-00042",
      "product": { "id": "0197e4a0-0000-7000-8000-000000001004", "sku": "P-1004", "name": "ドライブシャフト" },
      "quantity": 250,
      "dueDate": "2026-09-18",
      "status": "InProgress",
      "isOverdue": true,
      "updatedAt": "2026-09-20T01:12:03Z"
    }
  ],
  "total": 87,
  "page": 1,
  "pageSize": 20,
  "sort": "dueDate",
  "dir": "asc"
}
```

Error codes

Code	Meaning	HTTP status
MSG-E015	Order-number fragment longer than 20 characters	400
MSG-E016	Malformed <code>dueFrom</code> or <code>dueTo</code>	400
MSG-E017	<code>dueFrom</code> after <code>dueTo</code>	400
MSG-E018	Unknown status value	400
MSG-E002	<code>productId</code> unparseable or not an existing product	400
MSG-E019	Unsupported <code>sort</code> , <code>dir</code> , <code>page</code> or <code>pageSize</code>	400

Code	Meaning	HTTP status
—	Not signed in	401
—	Missing role (client shows MSG-E020)	403
MSG-E013	Unexpected failure	500

Every 400 returns `errors` keyed by the offending parameter name, so the filter panel can attach each message to its own field. Multiple invalid parameters are reported together in one response rather than one at a time.

2. GET /api/products

Unchanged — designed in 001_DD-API §1 and already implemented. SCR-002 calls it for the product filter options (002_BD FN-012) and uses the same "{sku} — {name}" label and SKU ordering as SCR-001 (WI-002 DEC-022). No paging or search is added: the list is 30 rows.

Unresolved decisions

None. The decisions this contract rests on are recorded in `work-items/WI-003/decisions.md` (DEC-002 paging and sort, DEC-005 multi-select status, DEC-010 fragment matching) and `work-items/WI-002/decisions.md` (DEC-011/DEC-017 plant clock, DEC-020 CSRF, DEC-022 product labels, DEC-023 the RFC 9457 error contract this reuses).

One limitation is recorded rather than decided here: the endpoint is gated by the existing `ProductionOrderEditor` policy, so a hypothetical read-only role cannot be expressed yet. That waits on WI-001 DEC-015 (the full role/permission matrix), which stays open.